Projectory | Milestone 3

APPENDIX I — PART 1: PROJECT STATUS AND UI REVIEW

Section: CSC 648-06 | Team: 03 (Projectory)

1. UI and Functionality Feedback (P1 Features Only)

Student Demonstration and Review Context

The team demonstrated the Projectory portfolio workflow, 3D model viewing and CAD-compatible STL mesh export, and AI chatbot functionality. The instructor indicated that the M3 grade/status was acceptable, while requesting a more focused and credible AI use case and a more interactive, collaborative experience for 3D projects. This statement records team-reported feedback; it is not a claim that every P1 requirement was independently verified on the deployed server.

Instructor's Comments on the Demo

AI scope and accuracy. The instructor questioned how reliably a broadly framed career-guidance chatbot could advise students, since career recommendations can be vague and difficult to substantiate. The suggested direction was to narrow the task toward academic and project-development guidance informed by educational sources, the student's profile and portfolio, and the conversation. Concrete suggestions could include relevant courses, campus events, hackathons, and nearby learning opportunities. The team also discussed whether 3D project artifacts could inform those suggestions.

3D project interaction and collaboration. The instructor wanted 3D models to be more than isolated uploads or exports. Students should be able to view and engage with one another's work, provide likes and comments, and see successive project/model iterations and descriptions of changes. The team should explore ways to make the development process visible and collaborative, rather than treating the 3D viewer as a static showcase.

Team's Action Plan Based on Feedback

Area	Planned action	Timing / verification
Grounded AI guidance	Narrow the guidance experience to course, skill, project and opportunity suggestions. Retrieve information from identified university/course/event sources and display citations, dates and uncertainty rather than unsupported assertions.	M4: demonstrate one source-backed recommendation flow; M5: verify relevance and error handling.
Personalization	Use authorized student profile fields, portfolio project metadata and user-provided conversation context. Let students review what information is used; do not infer mastery solely from a 3D scan.	M4: demonstrate project-informed results; M5: test different student profiles.
Social project engagement	Add basic project likes and comments, with authenticated identity, permission checks and a clear UI for viewing feedback.	M4: basic end-to-end interaction; M5: regression and abuse/permission checks.
Design iterations	Allow multiple named model/project revisions with timestamps and short change notes. Show revision history	M4: revision upload and display; M5: test version retrieval and downloads.

	in the project page; reserve automated geometry diff for later work.	
Integration and quality	Verify AI frontend/backend contract, actual provider use, deployed storage, model access, visibility, and regression flows; continue reviewed feature PRs into dev.	Throughout M4–M5; maintain reproducible test evidence.

2. List of P1 Features Committed for M5 Delivery

Final P1 feature commitments for Milestone 5. The following scope is recorded as the team's M5 delivery commitment.

Committed P1 feature	M5 acceptance target
Account authentication and authorization	Register, log in/out, validate input and restrict modifications to the authenticated owner.
Student profile and education history	Create, view and edit biography, interests, education, skills and experience, with saved changes displayed in the portfolio.
Portfolio and project management	Create, edit, view and discover projects; publish/unpublish portfolios and enforce authorized project access.
Focused AI academic guidance	Provide profile- and project-informed recommendations with source-backed course/learning suggestions, transparent reasoning, and clear provider/error states.
Recommendation results	Display strengths, development areas, and concrete next steps, distinguishing retrieved source facts from generated suggestions.
Basic collaborative project feedback	Allow authenticated likes/comments on projects and display feedback to viewers.
Project/model revision history	Upload and display multiple versions of a project/model with timestamps and student-written change summaries.

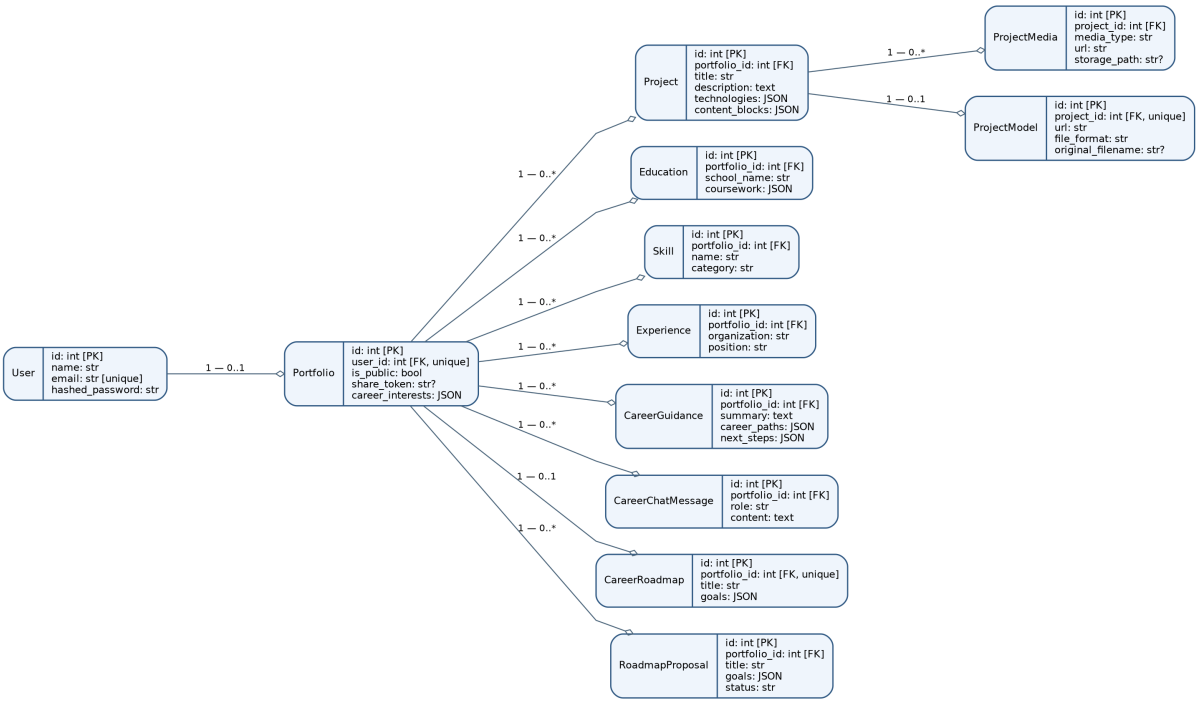
Scope boundaries: Existing GLB viewing and STL export remain useful demonstrable functionality; STL is a CAD-compatible triangle mesh, not parametric CAD. Automatic geometric comparison, real-time multiuser CAD editing, autonomous event scraping, and direct geometric skill assessment are not included in this proposed P1 list. Confirm any additions or removals with the instructor/team before freezing.

3. Architecture

Architecture deliverables: one high-level UML domain class diagram (main SQLAlchemy entities, selected attributes, and multiplicities) and six formal UML sequence diagrams tracing the implemented user workflows.

UML class diagram — implemented SQLAlchemy domain model

Class diagram — grounded in the SQLAlchemy ORM models in the uploaded dev branch.



UML sequence diagrams — six core application workflows

Formal UML sequence diagrams with participant lifelines, directional messages, return arrows, and annotated internal validation. Based on the uploaded dev-branch source.

Diagram 1. Registration and login

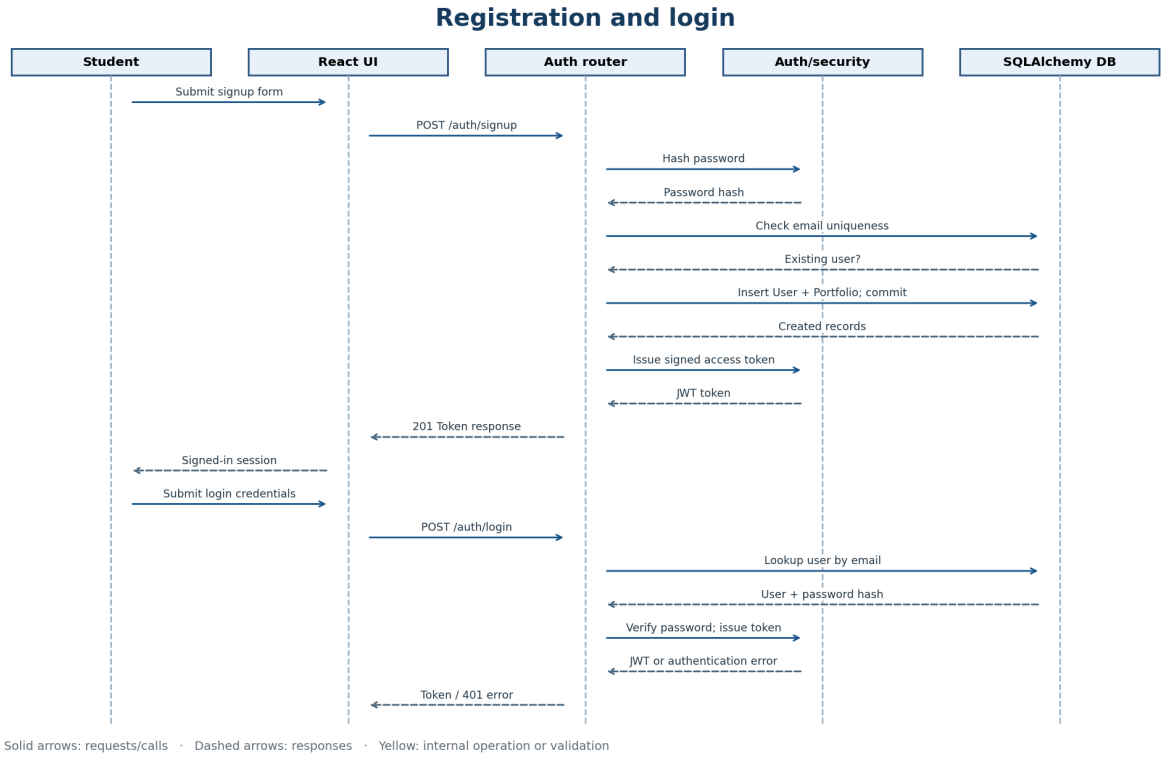
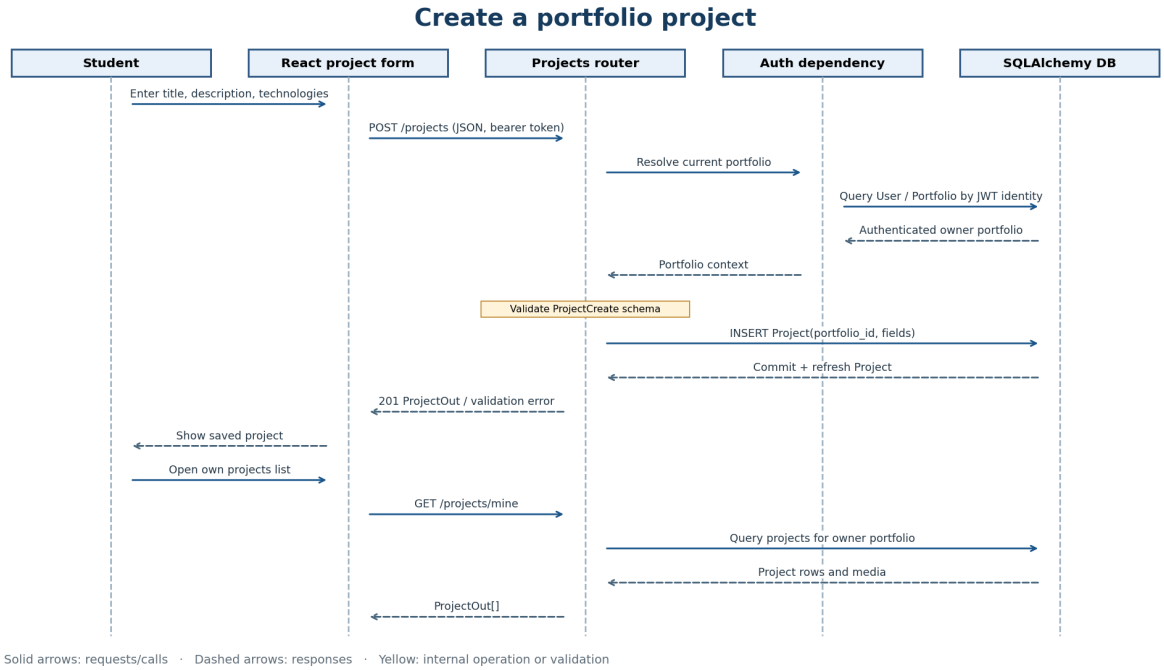
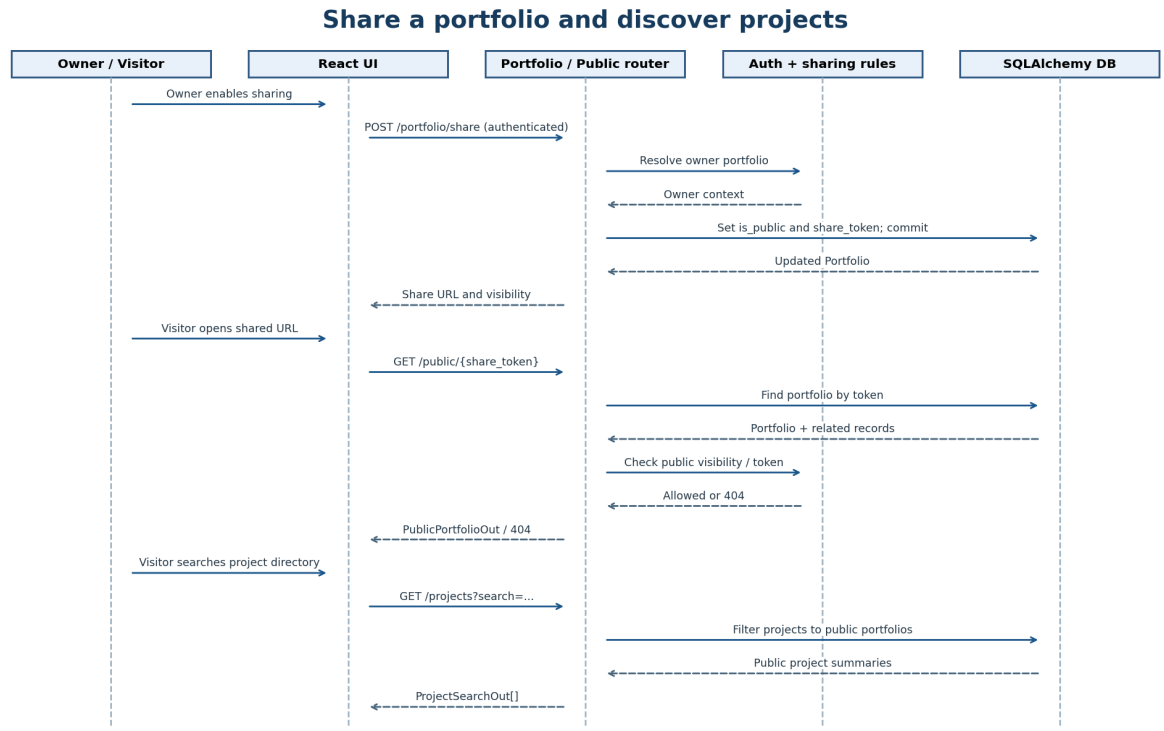


Diagram 2. Create a portfolio project



Solid arrows: requests/calls · Dashed arrows: responses · Yellow: internal operation or validation

Diagram 3. Publish and discover projects



Solid arrows: requests/calls · Dashed arrows: responses · Yellow: internal operation or validation

Diagram 4. Upload and display a GLB model

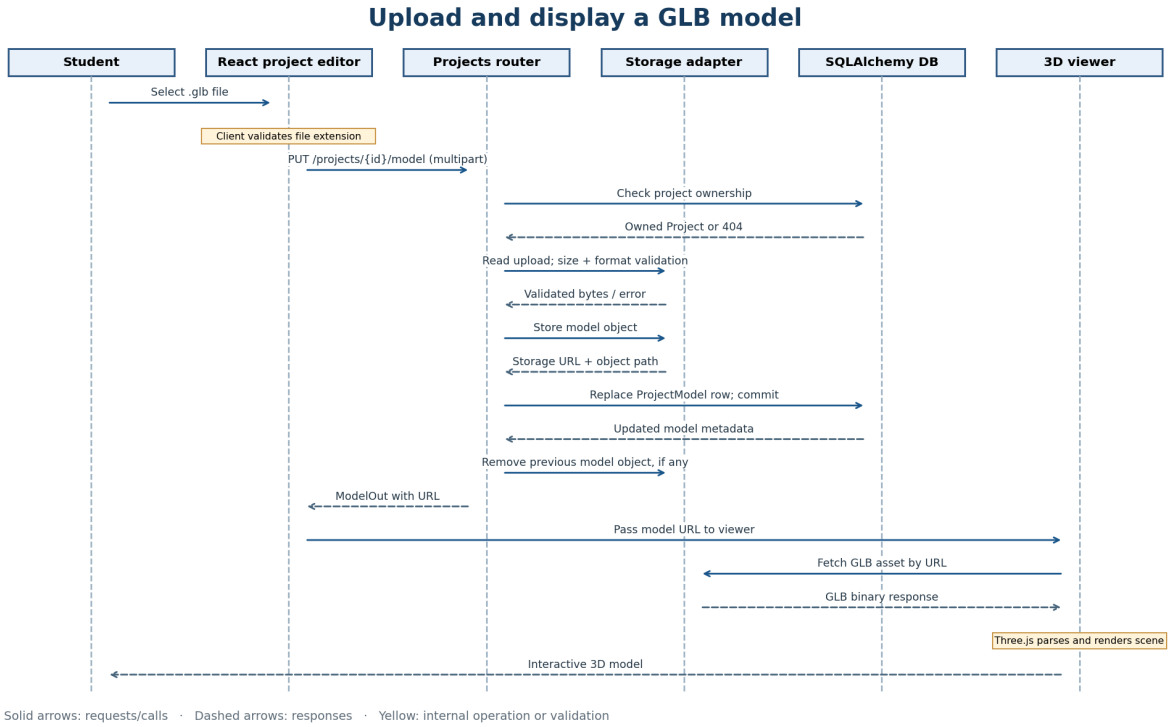
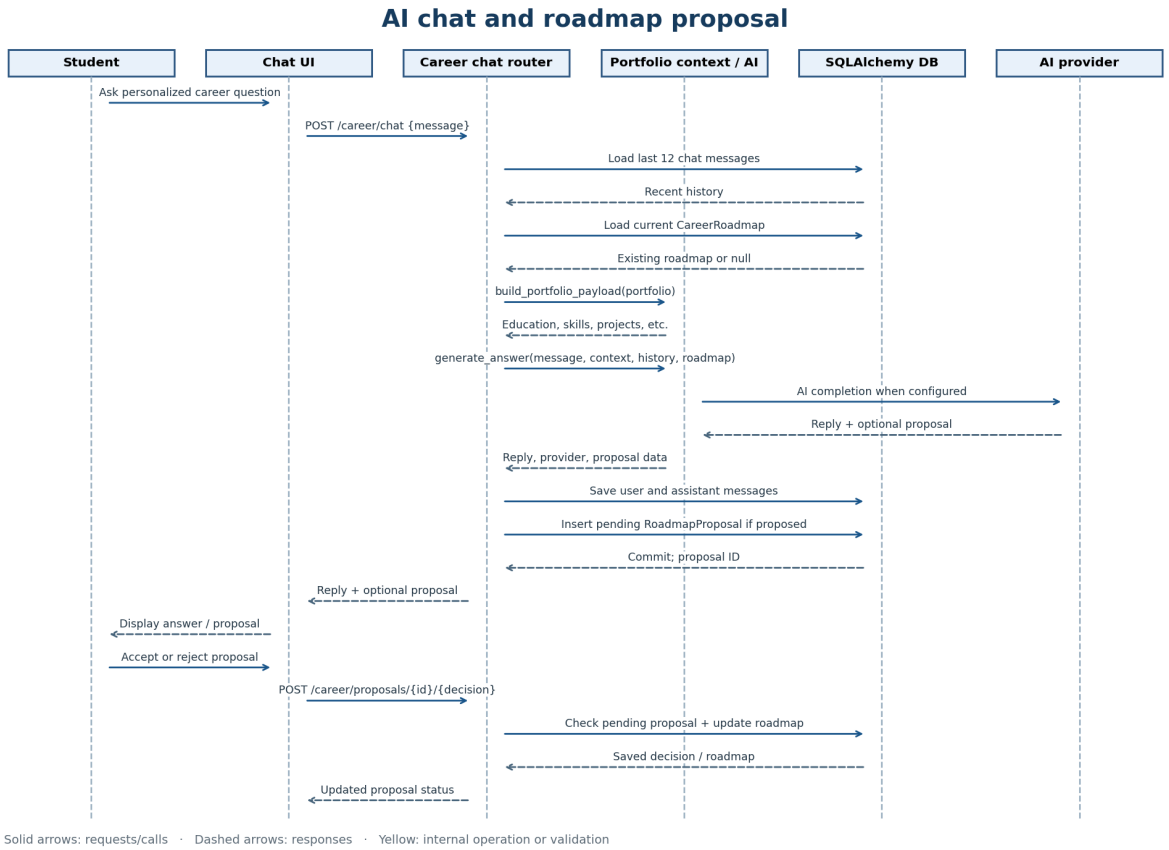
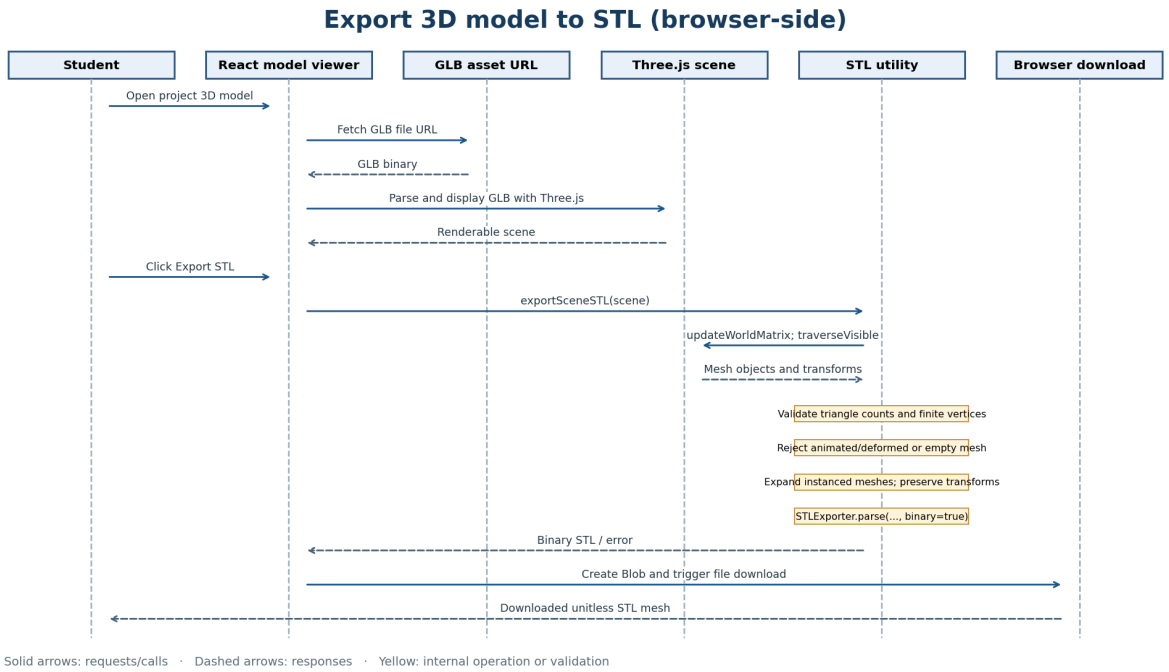


Diagram 5. AI career chatbot and roadmap proposal workflow



* The dev branch includes authenticated POST /career/chat, persistent chat history, optional roadmap proposals, and an approval/rejection endpoint. Real provider responses depend on runtime configuration.

Diagram 6. GLB-to-STL export (implemented in dev)



* The dev branch contains ModelViewer.jsx and the STL conversion utility (stl.js). STL is a CAD-compatible triangle mesh, not a parametric CAD file.

Source and scope note: UML updated against the uploaded SFSU-csc648-06-Fall26-team03-dev.zip. Diagrams reflect source-code paths, not independently verified production deployment.

4. Project Status — Actual Risks and Mitigation

Observed risk	Evidence / impact	Mitigation
AI accuracy and relevance	Broad career recommendations are difficult to verify; current guidance may be insufficiently grounded.	Restrict claims to supported suggestions, use official sources and source dates, test relevance, and expose uncertainty.
AI integration	Earlier audit found chat UI/backend endpoint mismatch and unverified live inference; teammate subsequently reported local chatbot success.	Confirm merged route and genuine end-to-end provider response on the deployed application.
Branch integration and review	Overlapping frontend PRs and PR #22 approval/merge-base issues occurred during M3.	Use feature PRs into dev, substantive review, integration tests, and controlled promotion to main.
Private model-file access	Known static or public object URLs may remain accessible after a portfolio is unpublished.	Document distinction between discovery privacy and file access; evaluate signed/authenticated model delivery.
CAD mesh quality	STL has no defined physical units; import scale, watertightness and CAD application compatibility were not fully validated.	Label STL accurately; test import, units, orientation and mesh cleanup with representative scans.
Scope and schedule	Professor feedback expands possible work shortly before the M5 feature	Agree to small, testable P1 scope, assign owners and milestones; defer

	freeze.	complex geometry analysis and real-time collaboration.

Post-Review Implementation Detail and Response to Instructor Feedback

A. Making AI guidance narrower, more accurate and verifiable

The team will reposition the AI component away from open-ended predictions about a student's eventual career and toward concrete academic and project-development decisions. A recommendation should answer a specific question—such as which course could help a student develop a demonstrated project skill, which prerequisite may be needed, or which campus activity may provide relevant experience—rather than assert that a student is suited to a profession. Projectory will combine the student's explicitly provided academic interests, skills, educational history, portfolio descriptions and conversation context with retrieved educational information. For courses and events, the system should prioritize official university catalog and organizer pages, retain source URLs and retrieval dates, and identify when information is missing or may be outdated. The interface should distinguish factual source information from the model's interpretation, explain why each suggestion relates to the student's stated goals, and avoid inventing offerings, schedules, eligibility or prerequisites.

To make this practical by M4, the first demonstration should cover a narrow, reproducible scenario: select a portfolio project, request relevant learning recommendations, retrieve a small set of authoritative sources, and present a short list of explained options with links. The team will test recommendations against known catalog details and verify that the assistant handles missing sources, conflicting prerequisites and unavailable provider responses without fabricating answers. The student should remain responsible for checking enrollment availability and making academic decisions. Integration with 3D work will initially use the student's project description, tools, revision notes and declared objectives; automatic recognition of engineering proficiency from raw geometry is outside the validated scope.

B. Turning 3D projects into collaborative, evolving work

The second response is to make each project a place for interaction and visible iteration rather than only a static portfolio item. The first deliverable should allow authenticated students to view publicly shared projects, leave comments, and express appreciation through likes. Project owners should be able to upload subsequent model versions with a revision title, timestamp and a short explanation of what changed, why the change was made, and what feedback motivated it. A project detail page can show the current model and a chronological history of earlier versions, making the engineering process understandable to peers and instructors. Backend permission checks should protect private projects and ensure that only authorized users can edit or replace a project's own revisions.

For M4, the team should prioritize a reliable end-to-end flow: publish a project, receive a comment, upload a revised GLB, and show the change note and both versions in the UI. The existing STL export can remain available for supported model revisions, with a clear explanation that it produces a triangle mesh rather than editable parametric CAD. More sophisticated enhancements—such as side-by-side geometric overlays, spatial annotations on a model, automated shape differences or simultaneous CAD editing—should be considered only after the basic social and revision workflow is stable. This phased approach addresses the instructor's collaboration concern while avoiding an unrealistic commitment to build a full CAD platform before M5.

C. Connecting collaboration and guidance without overstating AI capabilities

The longer-term value of Projectory is the relationship between a student's work, the feedback it receives and the next learning opportunity. Once revision history and comments are available, the AI may summarize student-authored change descriptions and suggest courses, workshops or project exercises that relate to the stated challenge. For example, a student who documents an enclosure redesign for a Raspberry Pi project might receive suggestions about CAD modeling or fabrication resources, supported by actual course or event listings. Such recommendations must remain traceable to the student's text and retrieved sources; a rendered 3D model by itself is not reliable evidence of design intent, mastery

or career readiness. The team will validate this combined workflow only after the individual guidance and collaboration features pass their own functional and access-control tests.

D. Acceptance criteria and delivery discipline

The team will translate the instructor's feedback into specific acceptance tests, assign owners, and track the finalized P1 commitments in the repository. M4 should show at least one real, source-grounded guidance request and one collaborative project iteration on the deployed system; M5 should focus on robustness, permissions, clear UX, and reproducible regression results. Each feature will be developed on a focused branch, reviewed through a pull request into dev, tested with the integrated frontend and backend, and promoted to main only after the release is verified. Any requirement that depends on third-party availability, live event listings or external model credentials will have an explicit fallback or clear unavailable state, not a fabricated success message.